



4.8 Spearman's rank correlation

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Spearman's rank correlation coefficient $r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$ measures the strength and direction of a MONOTONIC (not necessarily linear) relationship between two variables, using the differences d between each pair's ranks and the number of pairs n . It is used when data is ordinal, when the relationship may be non-linear but still consistently increasing/decreasing, or to reduce the influence of outliers; it is interpreted on the same -1 to 1 scale as Pearson's r .

Worked example: follow the reasoning

Two judges rank 5 contestants: Judge 1 gives ranks 1,2,3,4,5 and Judge 2 gives ranks 2,1,4,3,5 (in the same contestant order). Find Spearman's rank correlation coefficient.

Solution: $d = -1, 1, -1, 1, 0$; $d^2 = 1, 1, 1, 1, 0$, so $\sum d^2 = 4$. $n = 5$. $r_s = 1 - \frac{6(4)}{5(25-1)} = 1 - \frac{24}{120} = 0.8$.

Quick check · multiple choice

Spearman's rank correlation is particularly useful when:

A Data is linear and continuous

B Data is ordinal or the relationship may be non-linear but monotonic

C Sample size is very large

D Variables are independent

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Seven café reviewers rank 7 coffees for taste (1=best) and separately rank them by price (1=cheapest). The rank differences (taste rank – price rank) for the 7 coffees are: 1, –2, 0, 3, –1, –2, 1. (a) Calculate $\sum d^2$. (b) Calculate Spearman's rank correlation coefficient, r_s , to 3 significant figures. (c) Interpret the value of r_s found in (b) in context. [6]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) d^2 : 1,4,0,9,1,4,1; $\Sigma d^2=20$. (b) $n=7$: $r_s=1-6(20)/(7(49-1))=1-120/336\approx 0.643$. (c) $r_s\approx 0.643$ indicates a moderate positive monotonic association: coffees ranked tastier tend also to be ranked more expensive, though the relationship is not very strong.

Quick check answer

B. Spearman's coefficient only needs ranked (ordinal) data and detects any monotonic relationship, not just a linear one.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

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